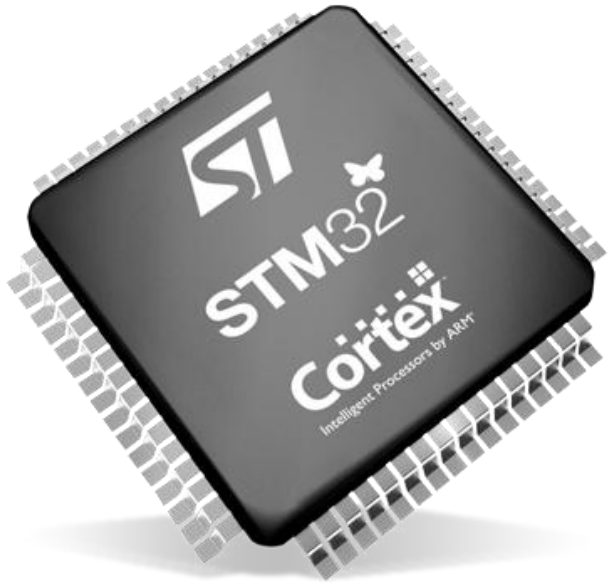


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KHOA ĐIỆN TỬ - VIỄN THÔNG

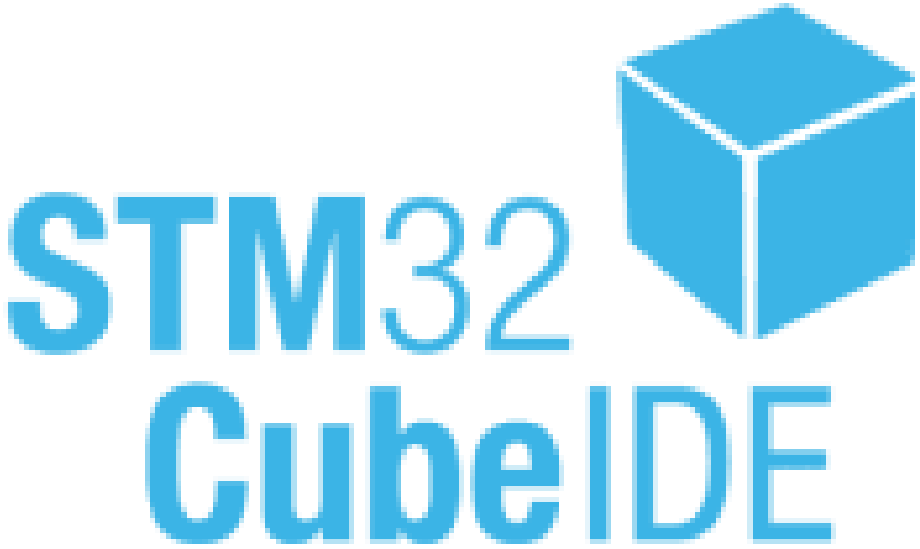
GIỚI THIỆU MÔN HỌC

- Môn học: Thực Hành Vi điều khiển
- Khoa Điện tử - Viễn thông
- Khóa 2023 – Chương trình CLC
- Giảng viên: CN. Nguyễn Vũ Minh Thành
- Email: nvmthanh@hcmus.edu.vn

Nội dung môn học

- 10 bài thực hành về lập trình STM32
- Tài liệu thực hành bản cứng
- Hình thức đánh giá:
  - Điểm danh: 20%
  - Bài tập hàng tuần: 40%
  - Thi cuối kỳ: 40%
- Phần mềm: STM32CubeIDE
- Bo thực hành: Bo mạch VĐK Grape32 Unleashed Kit
- Nội dung cần chuẩn bị:
  - Tạo tài khoản myST
  - Tải phần mềm STM32CubeIDE
    - Tạo project đầu tiên sử dụng chip STM32F103RCT6

STM32CubeIDE - Integrated Development Environment for STM32

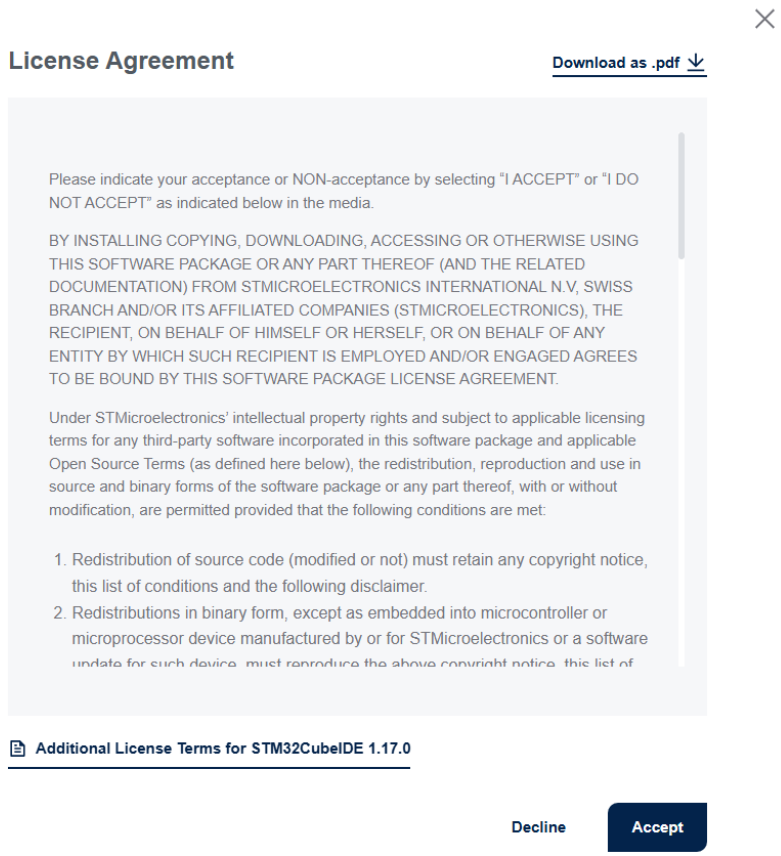


Get Software

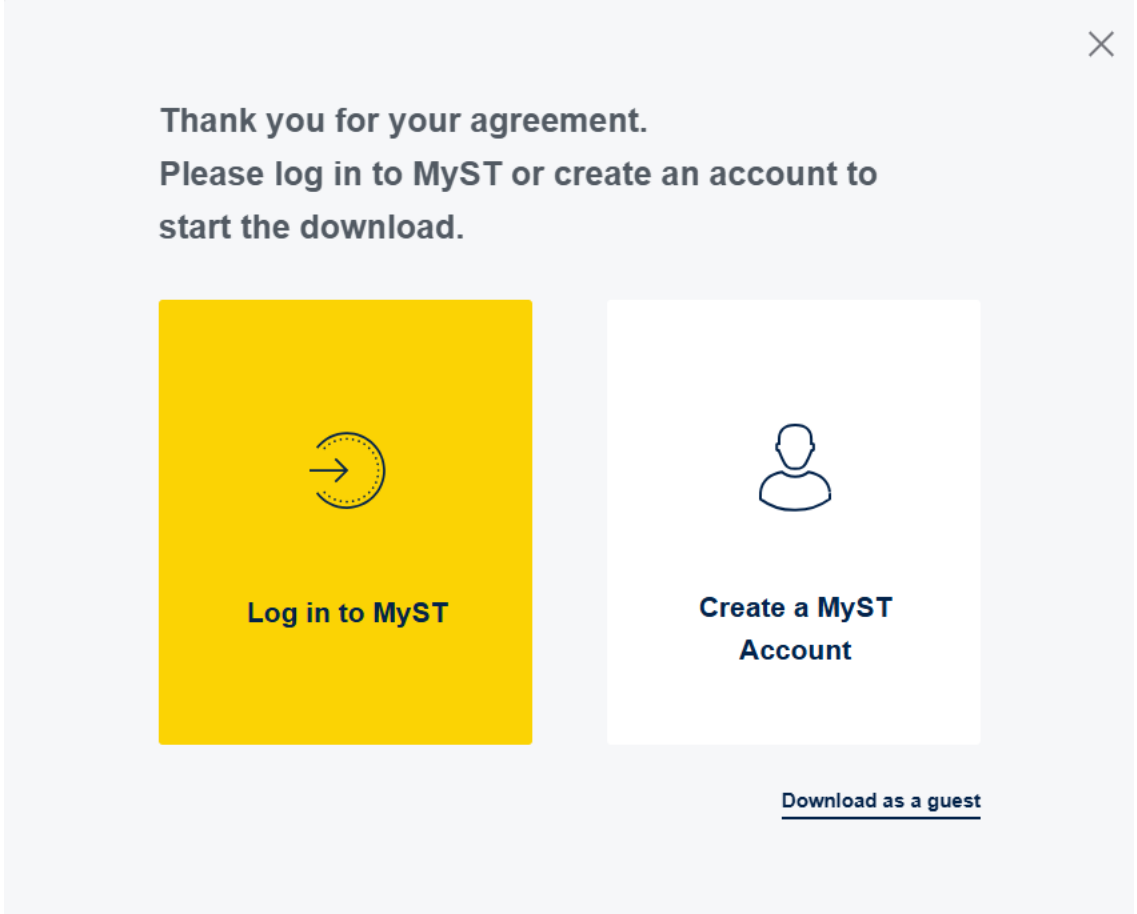
	Part Number	General Description	Latest version	Download	All versions
+	STM32CubeIDE-DEB	STM32CubeIDE Debian Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Lnx	STM32CubeIDE Generic Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Mac	STM32CubeIDE macOS Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-RPM	STM32CubeIDE RPM Linux Installer	1.17.0	Get latest	Select version
+	STM32CubeIDE-Win	STM32CubeIDE Windows Installer	1.17.0	Get latest	Select version

- Link: STM32CubeIDE - Integrated Development Environment for STM32 - STMicroelectronics

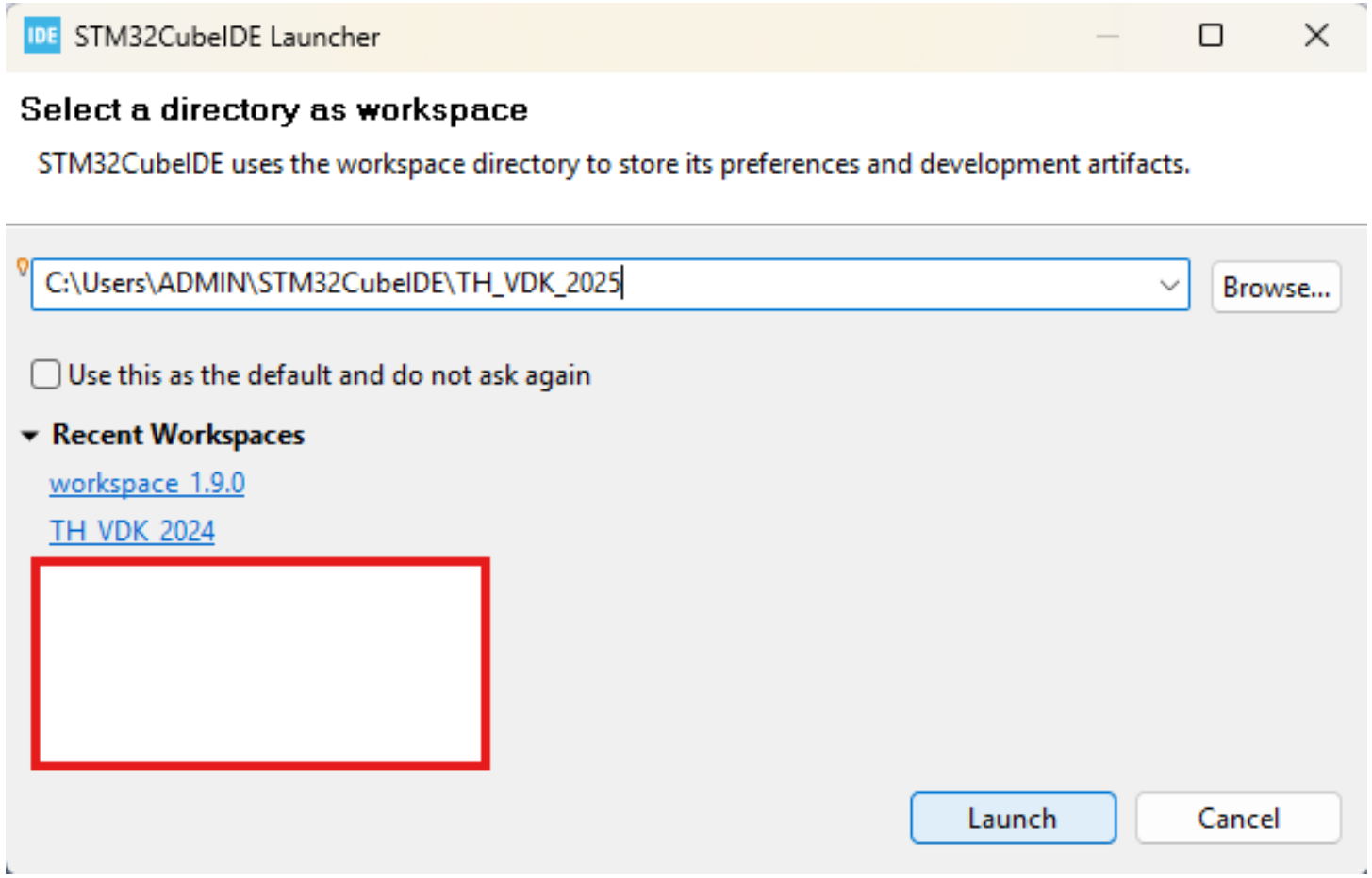
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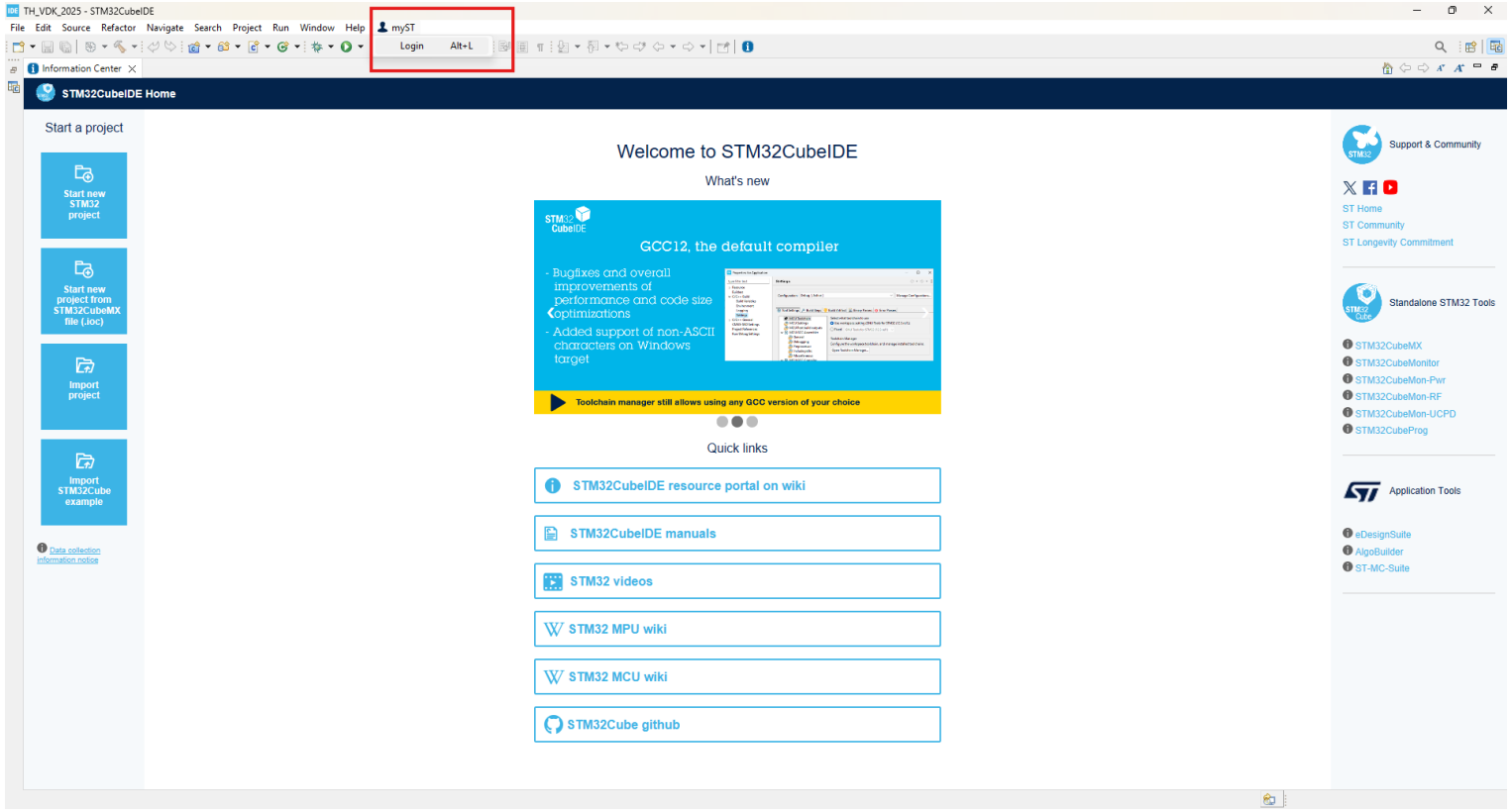
Start the download



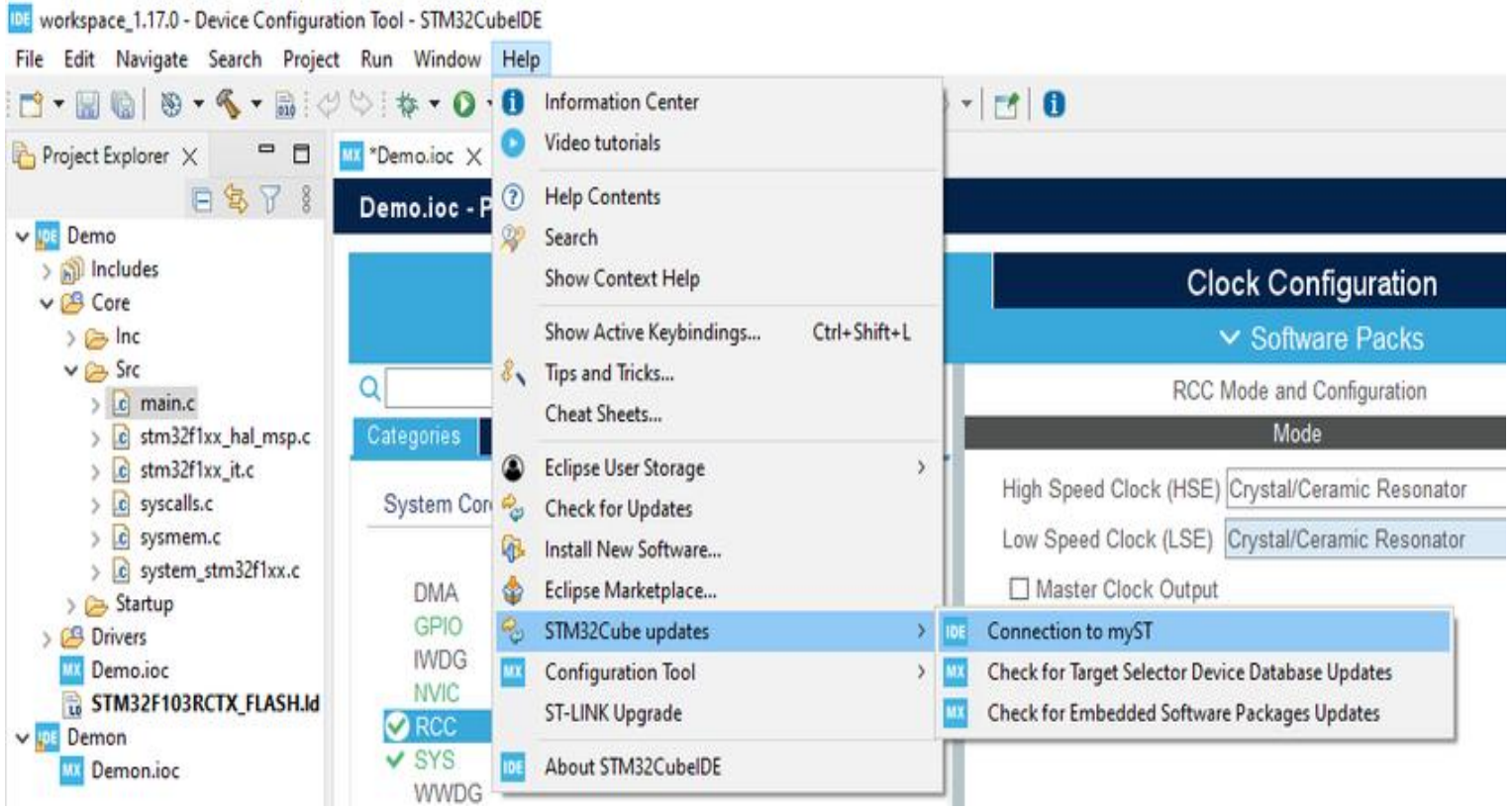
STM32CubeIDE Launcher



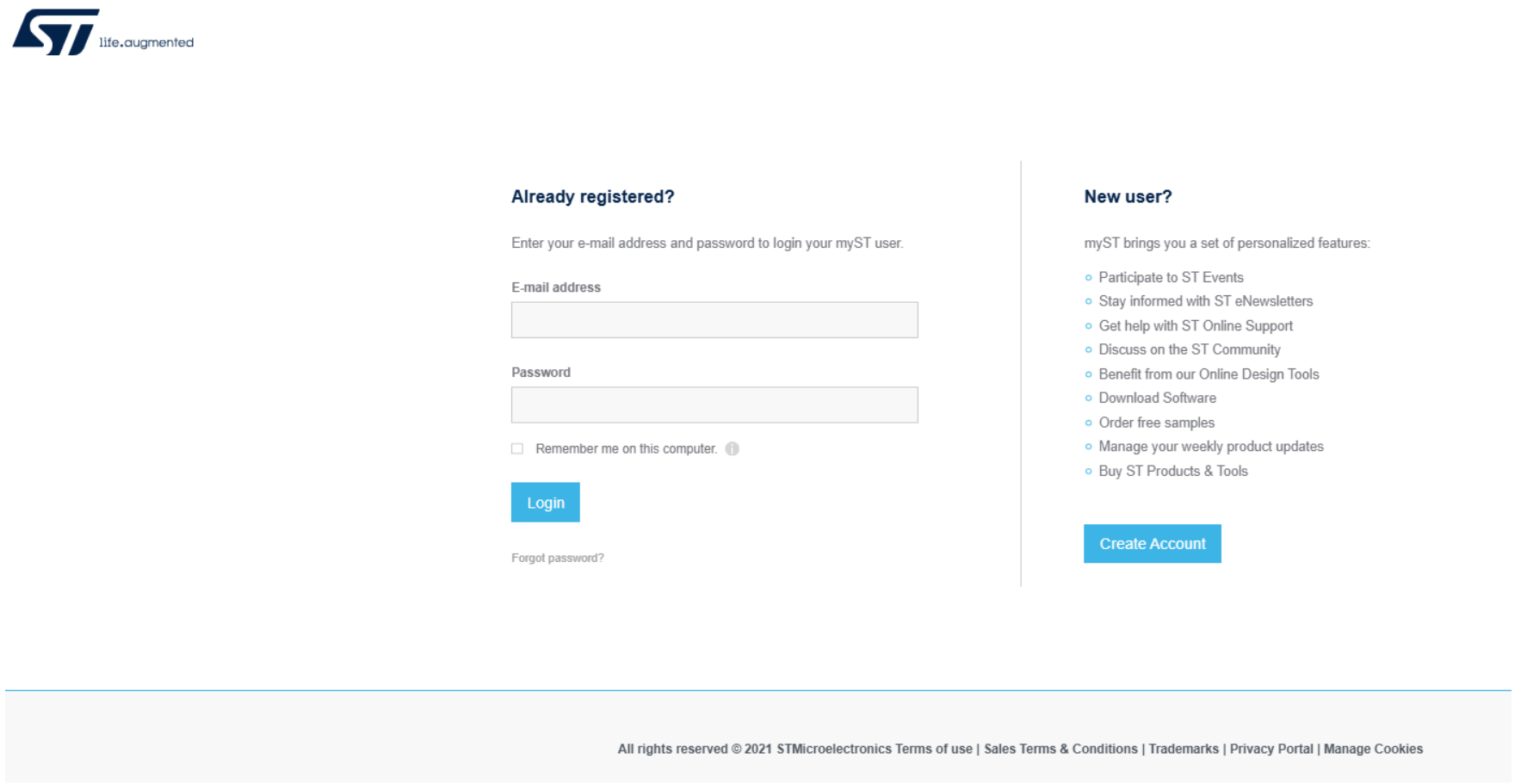
STM32 Cube IDE – Information Center & Login



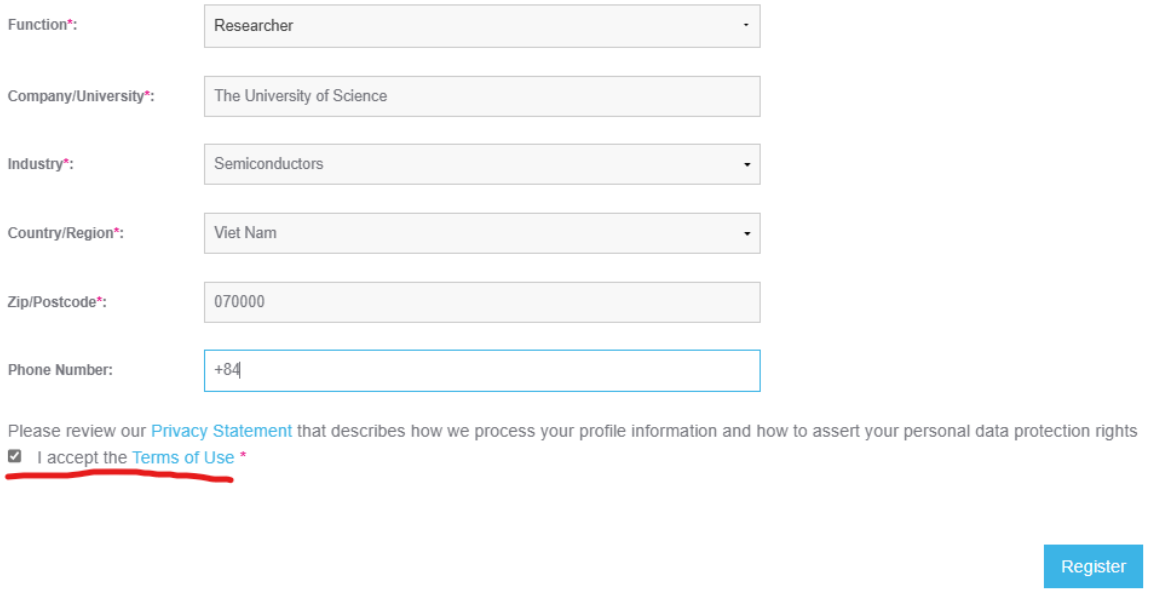
STM32 Cube IDE – Information Center & Login (New Version)



ST Account Login



ST Account Registration





ST Account Registration

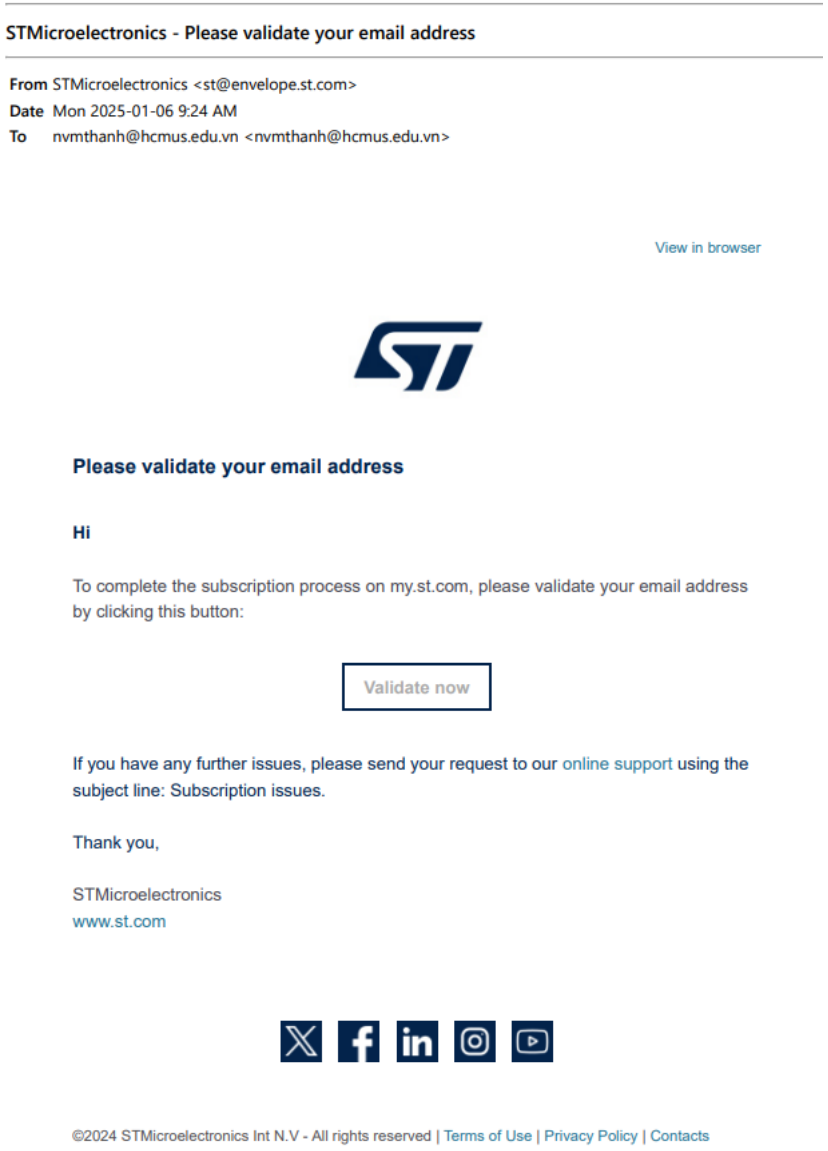
Thank you

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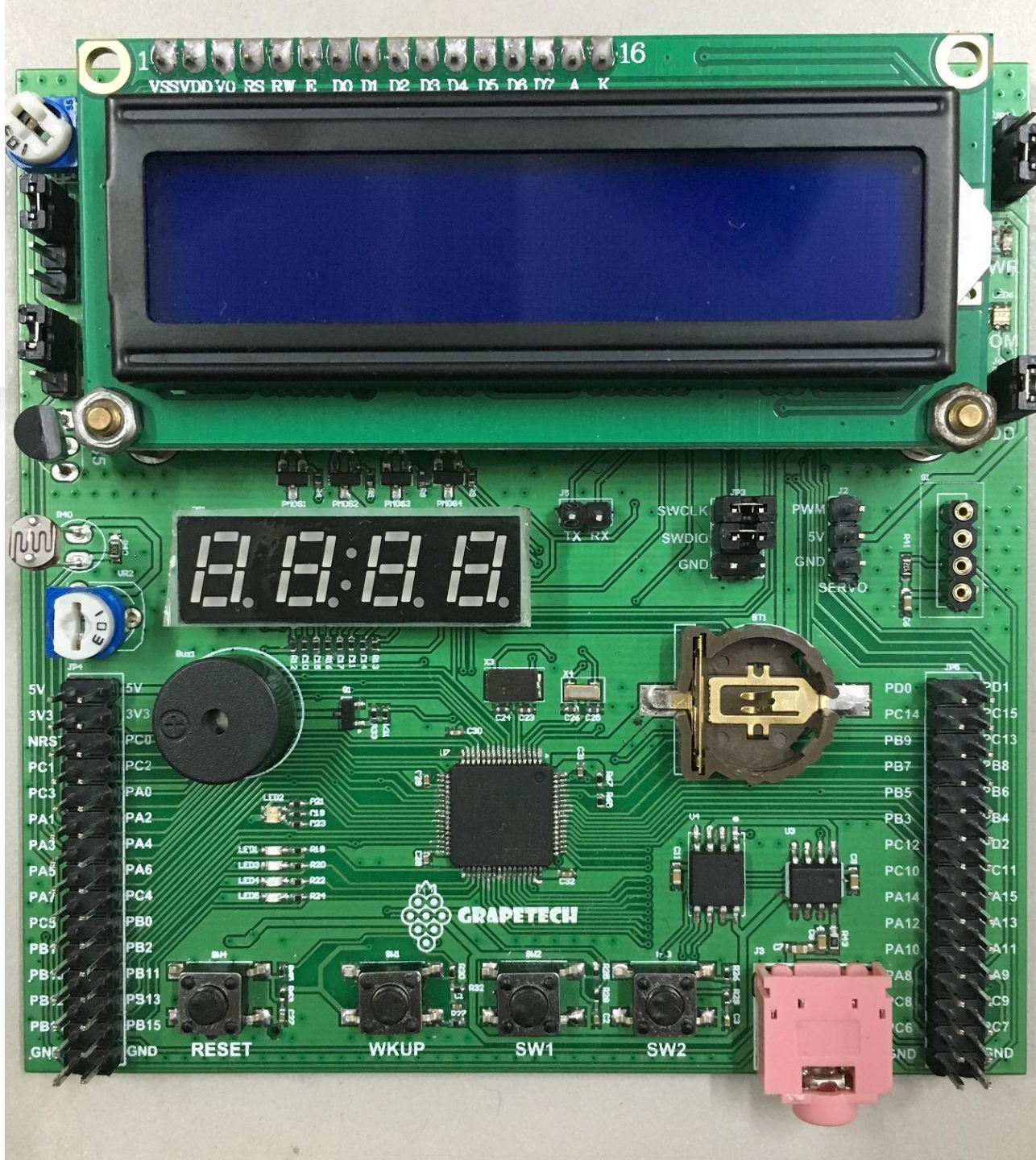
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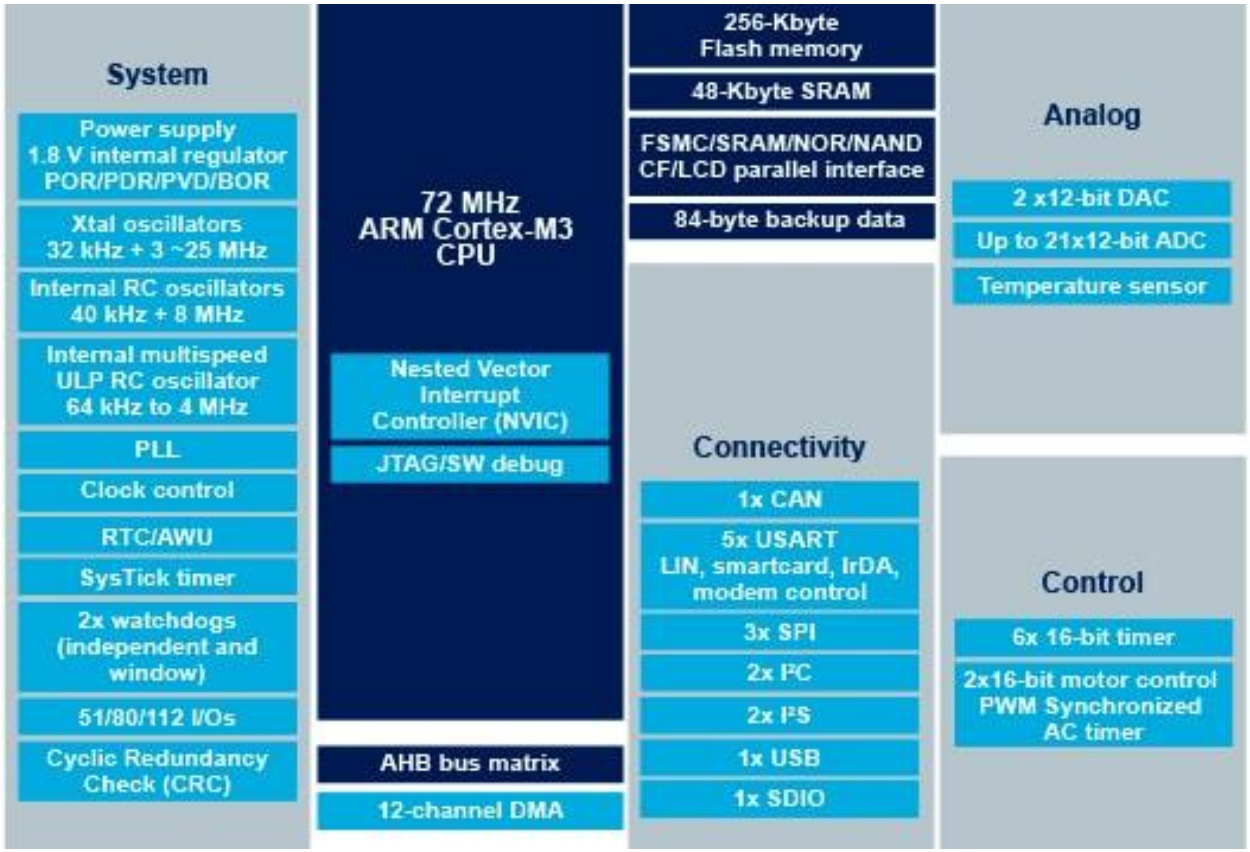
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Bo mạch VĐK Grape32 Unleashed Kit

- Controller: STM32F103RCT6 - 256KB FLASH, 48KB RAM
- Thông số kĩ thuật:
  - Nguồn cấp trực tiếp qua cổng USB (5V)
  - Các ngoại vi cơ bản:
    - 4 x LED và nút nhấn trên board, 1 x LED RGB
    - 4 x LED 7 đoạn, LCD 16×2
    - Ngõ ra Servo cho phép nghiên cứu về tính năng PWM
  - Các cảm biến LM35, quang trở; chân cắm cho cảm biến DHT11.
  - Các IC chức năng như NAND flash 4MB, DAC 12-bit, DS1307, EEPROM.
  - Chức năng khác: hỗ trợ cắm thẻ nhớ MicroSD, còi buzzer, Line-out ngõ ra



STM32F103RC System architecture



STM32 Labs

GPIO – System Configurations

- BÀI 1: GPIO – LED & NÚT NHẤN
- BÀI 2: GPIO – LED 7 ĐOẠN
- BÀI 3: GPIO – LCD

ADC

- BÀI 4: ADC – Cảm biến nhiệt độ LM35 & Cảm biến ánh sáng

TIMER

- BÀI 5: TIMER – COUNTER
- BÀI 6: TIMER – PWM

INTERRUPT

- BÀI 7: NGẮT (INTERRUPT)

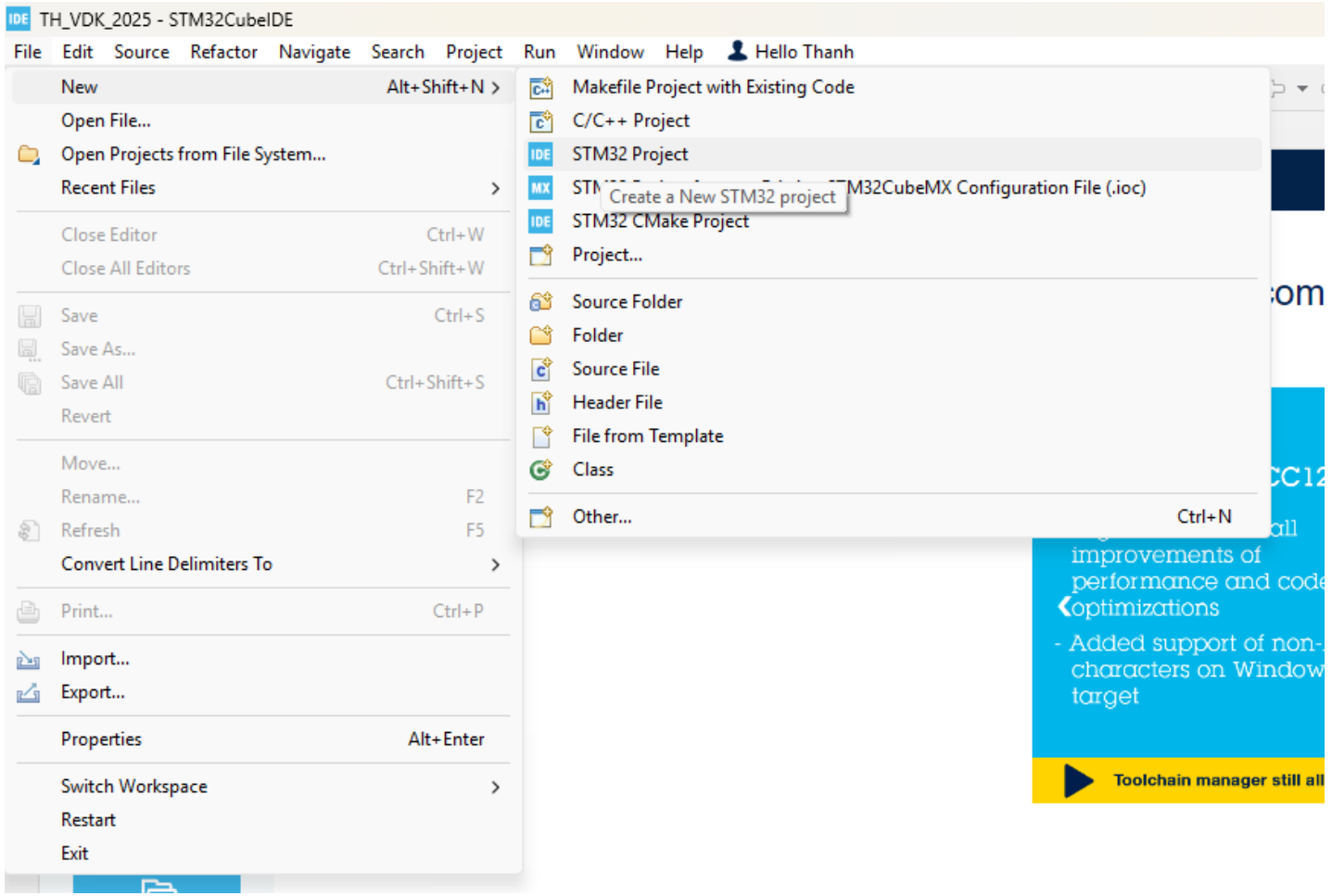
CONNECTIVITY

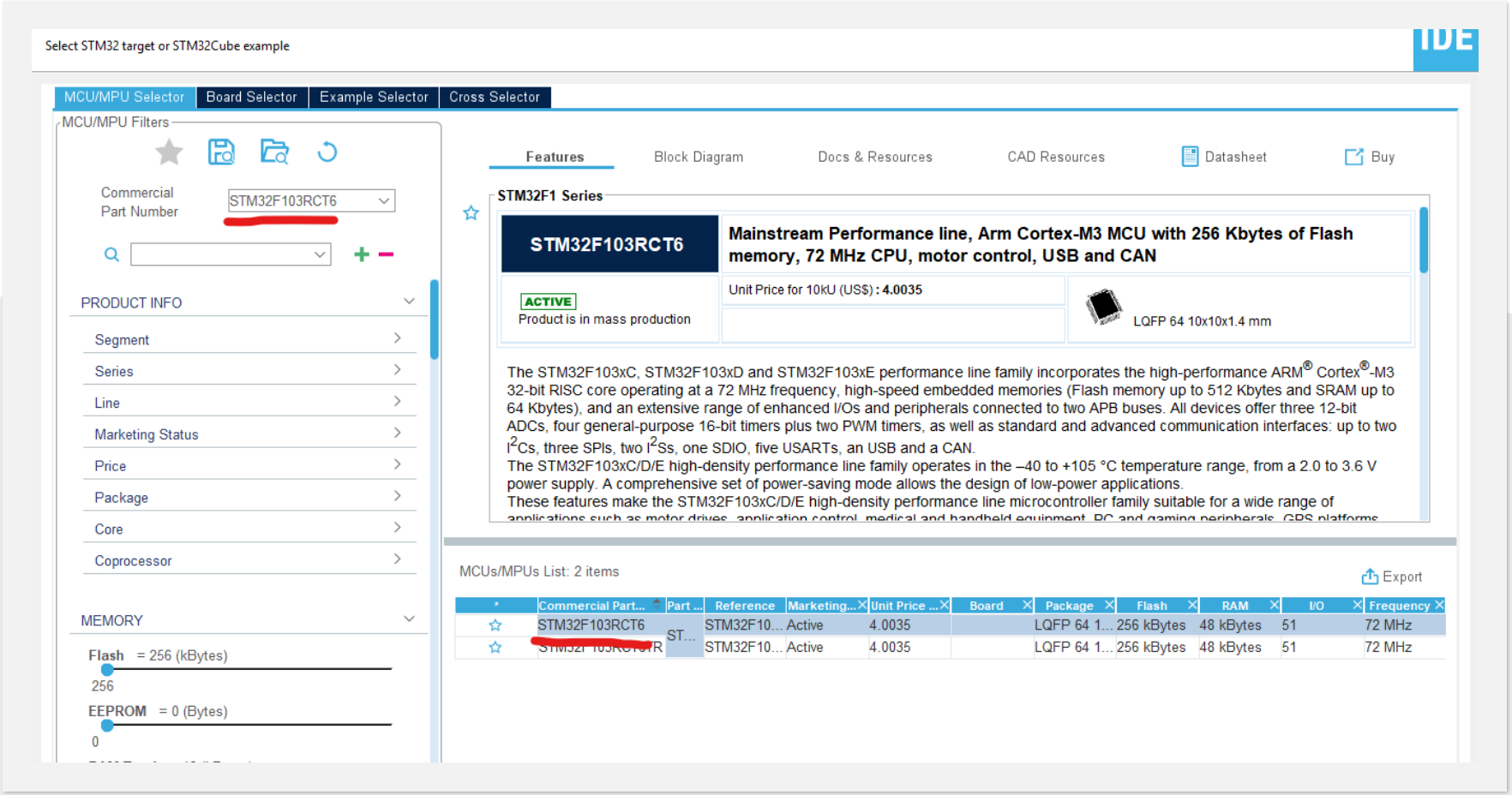
- BÀI 8: GIAO TIẾP UART
- BÀI 9: GIAO TIẾP I2C
- BÀI 10: GIAO TIẾP SPI

STM32 Workflows

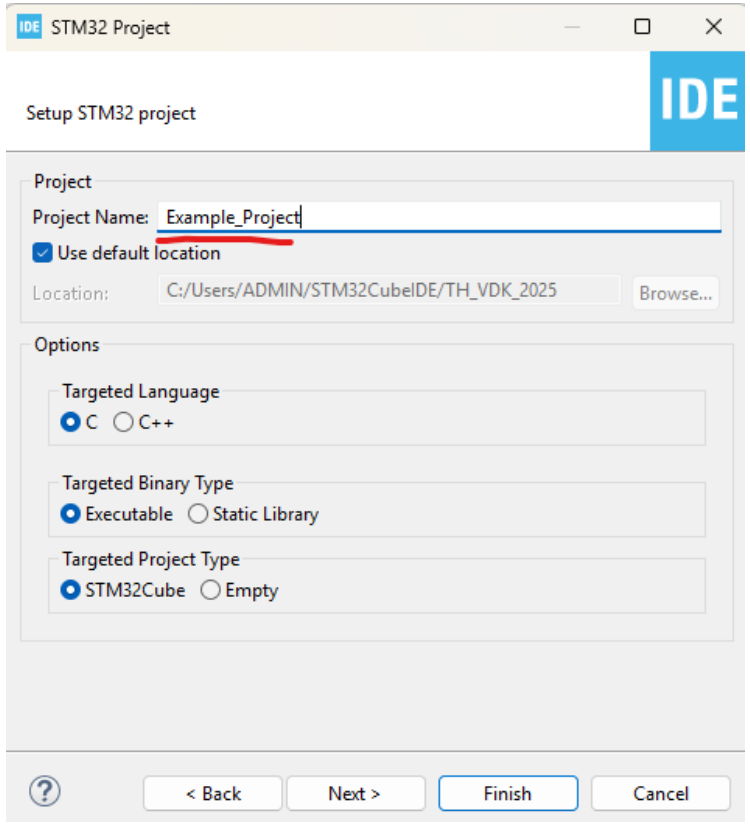
- Initialization
  - Create Project
  - Configure Chip Peripherals
  - Configure Debugger
- Main Operation
  - Code Generation
  - Coding & Compilation
  - Code Uploading
  - Functional Debugging

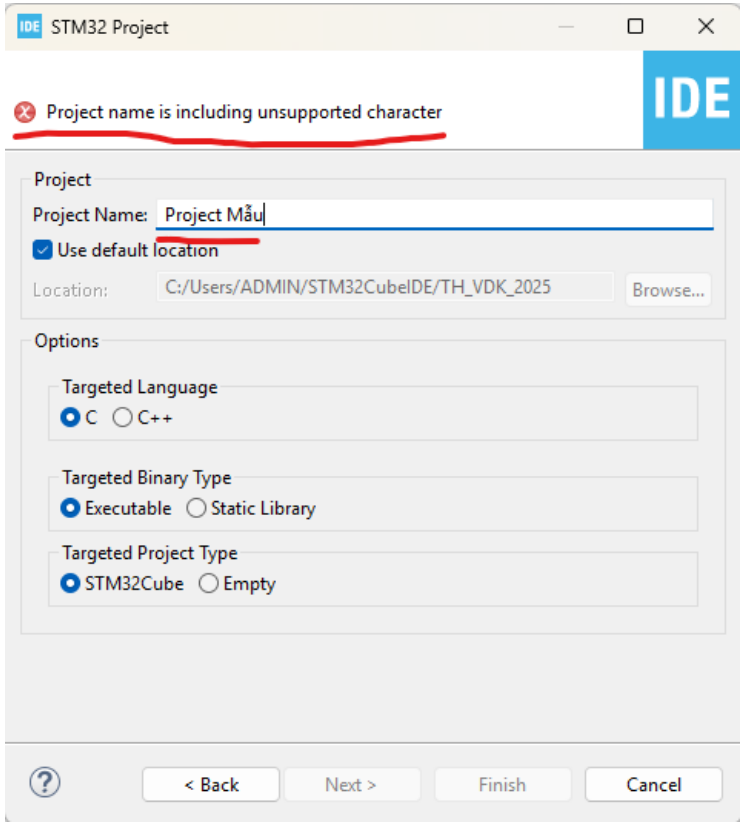
Tạo project trên STM32CubeIDE



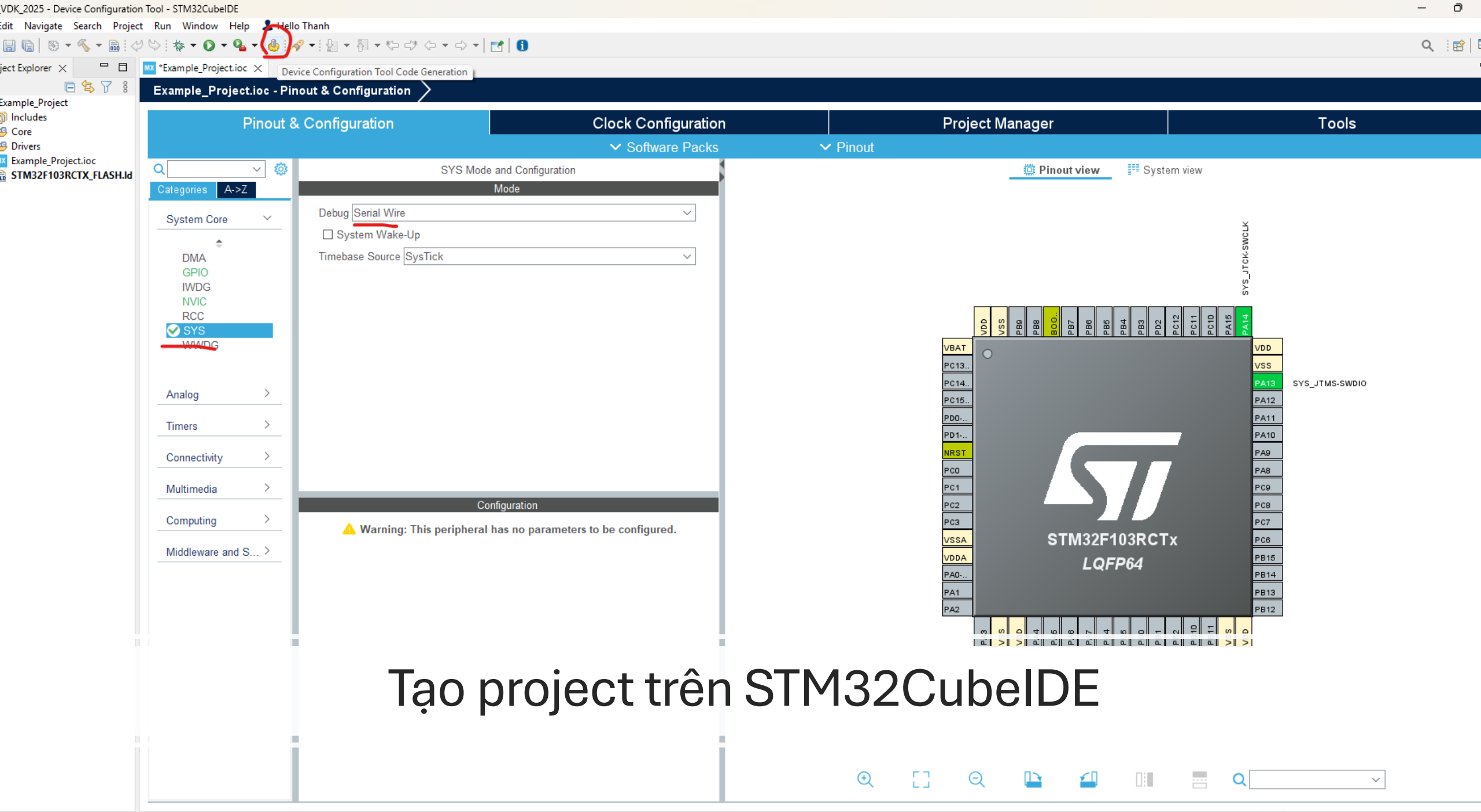


Tạo project trên STM32CubeIDE





Tạo project trên STM32CubeIDE


VDK_2025 - Device Configuration Tool - STM32CubeIDE
Edit Navigate Search Project Run Window Help Hello Thanh
Device Configuration Tool Code Generation
*Example_Project.ioc
Example_Project.ioc - Pinout & Configuration
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
Example_Project
Includes
Core
Drivers
Example_Project.ioc
STM32F103RCTX_FLASH.ld
Categories
A->Z
System Core
DMA
GPIO
IWDG
NVIC
RCC
SYS
WWDG
Analog
Timers
Connectivity
Multimedia
Computing
Middleware and S...
SYS Mode and Configuration
Mode
Debug
Serial Wire
System Wake-Up
Timebase Source
SysTick
Configuration
Warning: This peripheral has no parameters to be configured.
Pinout view
System view
SYS_JTCK-SWCLK
SYS_JTMS-SWDIO
STM32F103RCTx
LQFP64
Tạo project trên STM32CubeIDE

Code Format

```
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------
-*/
#include "main.h"

/* Private includes ----------------------------------------------------------
-*/

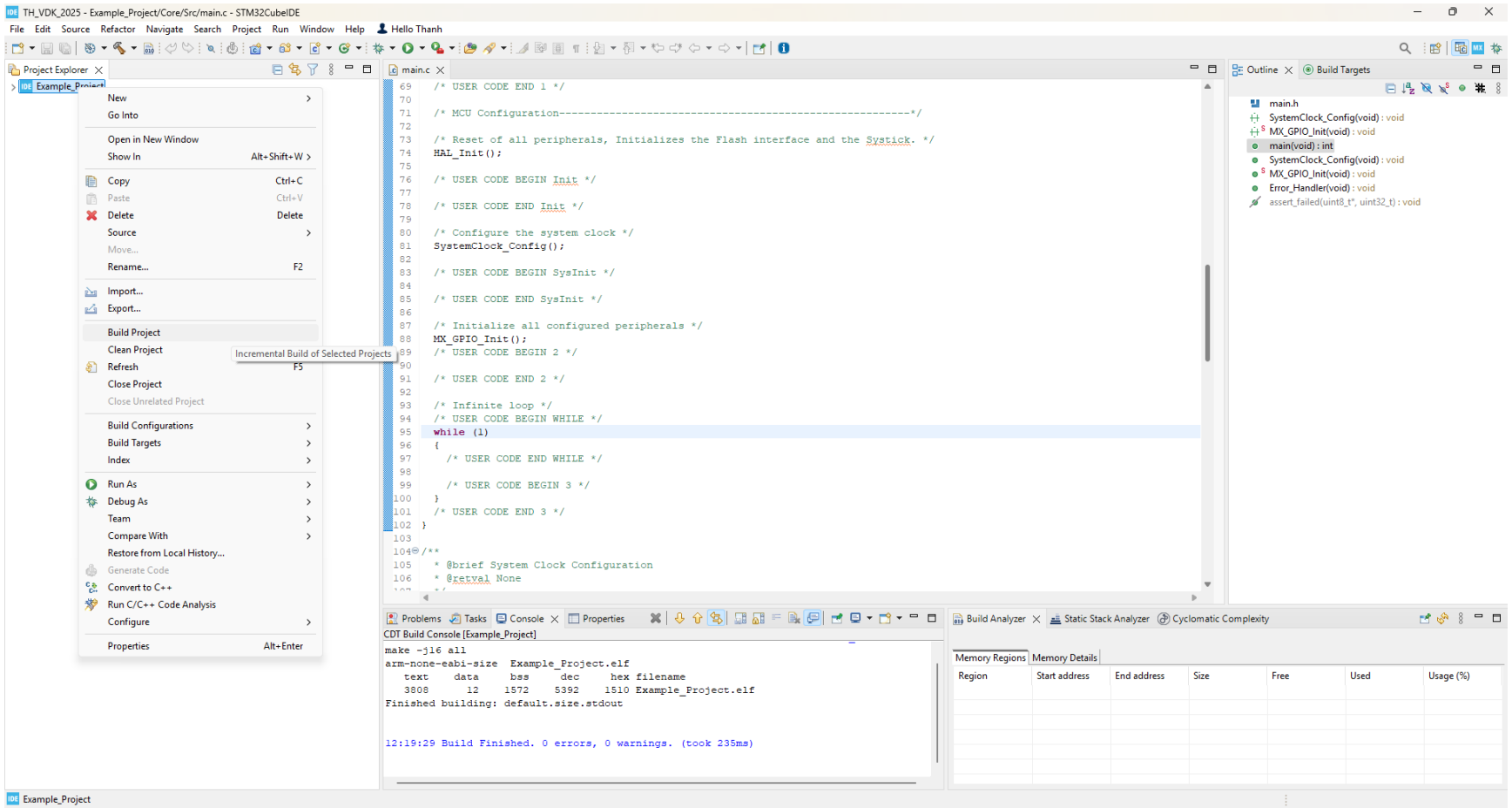
int main(void)
{

  /* MCU Configuration--------------------------------------------------------
-*/

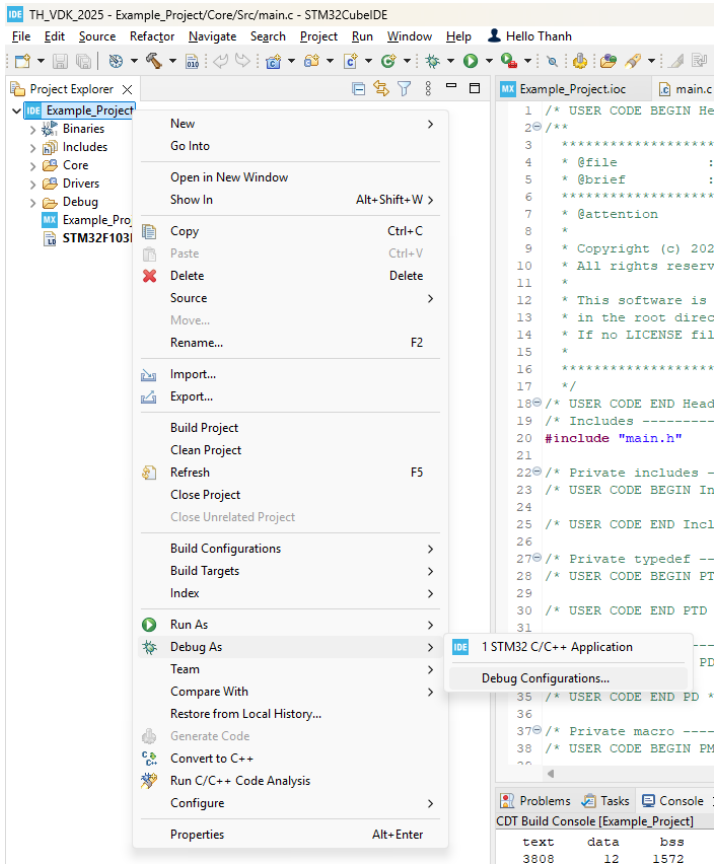
  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

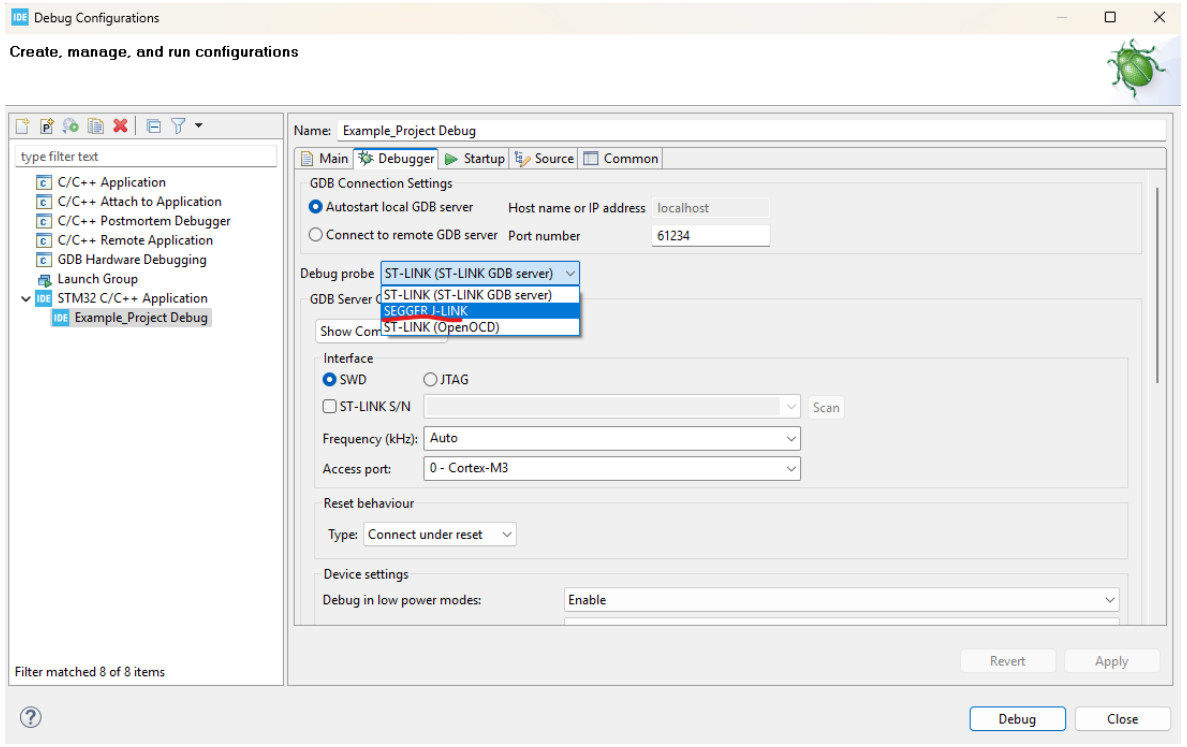
  }
}
```

Project Build

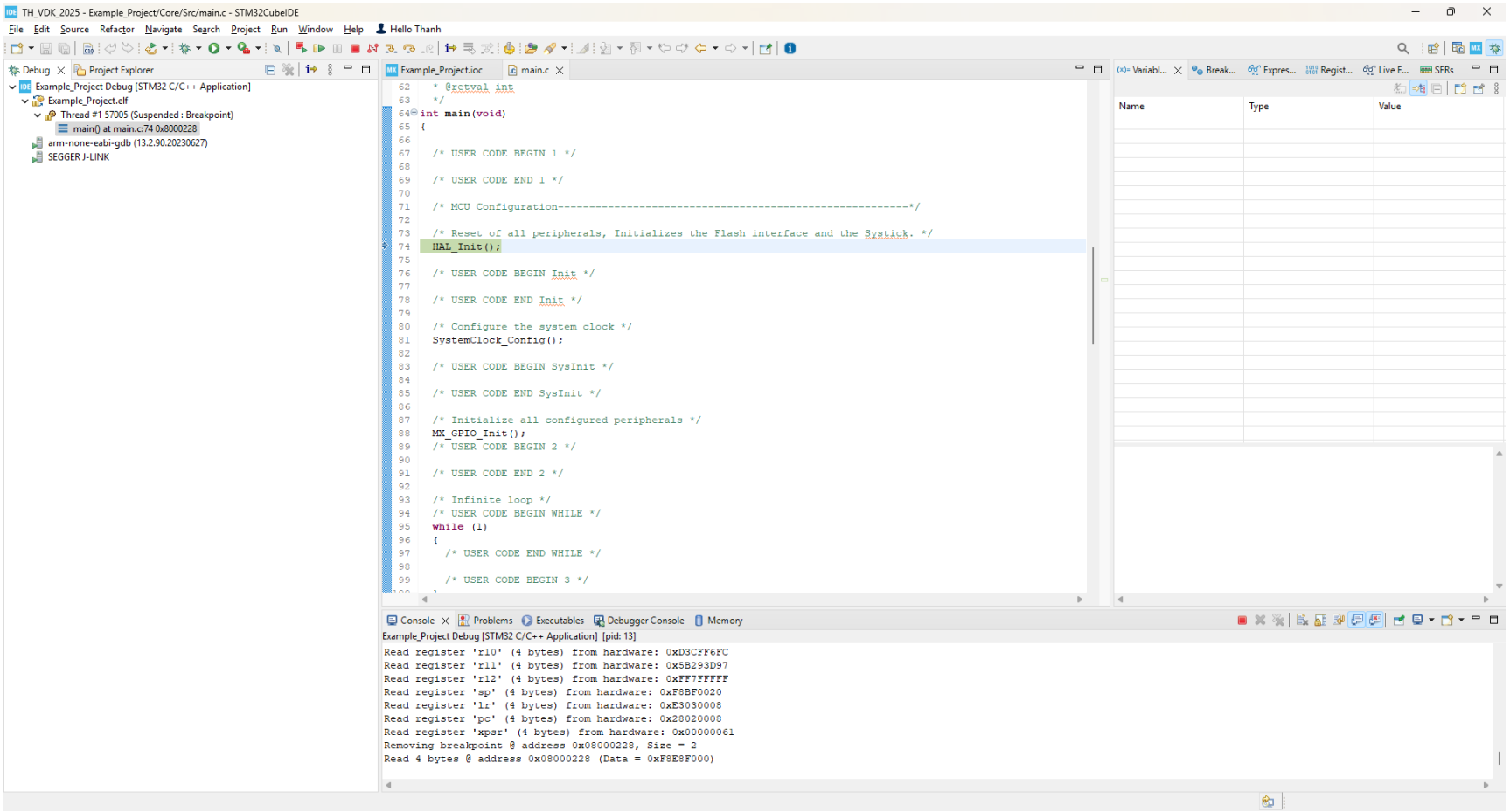


Debug Configuration





Debug Perspective



Programming Style

- Hardware Abstraction Layer (HAL – Môn TH)
  - `HAL_GPIO_WritePin(GPIOB, GPIO_PIN_12, GPIO_PIN_SET);`
  - `HAL_GPIO_ReadPin(GPIOB, GPIO_PIN_1);`
- Register-Level Programming (Môn LT)
  - `GPIOB->ODR |= (1 << 12);`
  - `uint8_t buttonState = !!(GPIOB->IDR & (1 << 1));`

Table 2. **HAL driver files**

File	Description
stm32f4xx_hal_ppp.c	Main peripheral/module driver file It includes the APIs that are common to all STM32 devices. *Example: stm32f4xx_hal_adc.c, stm32f4xx_hal_irda.c.*
stm32f4xx_hal_ppp.h	Header file of the main driver C file It includes common data, handle and enumeration structures, define statements and macros, as well as the exported generic APIs. *Example:stm32f4xx_hal_adc.h,stm32f4xx_hal_irda.h.*
stm32f4xx_hal_ppp_ex.c	Extension file of a peripheral/module driver. It includes the specific APIs for a given part number or family, as well as the newly defined APIs that overwrite the default generic APIs if the internal process is implemented in different way. *Example:stm32f4xx_hal_adc_ex.c,stm32f4xx_hal_flash_ex.c.*
stm32f4xx_hal_ppp_ex.h	Header file of the extension C file It includes the specific data and enumeration structures, define statements and macros, as well as the exported device part number specific APIs *Example: stm32f4xx_hal_adc_ex.h,stm32f4xx_hal_flash_ex.h.*
stm32f4xx_hal.c	This file is used for HAL initialization and contains DBGMCU, Remap and Time Delay based on SysTick APIs.
stm32f4xx_hal.h	stm32f4xx_hal.c header file
stm32f4xx_hal_msp_template.c	Template file to be copied to the user application folder It contains the MSP initialization and de-initialization (main routine and callbacks) of the peripheral used in the user application.
stm32f4xx_hal_conf_template.h	Template file allowing to customize the drivers for a given application
stm32f4xx_hal_def.h	Common HAL resources such as common define statements, enumerations, structures and macros

References

- Bo mạch VĐK Grape32 Unleashed Kit – Grapestore
  - Schematic: Grapini32_Grape32Unleashed_Boards/Schematics/Grape32Unleashed_schematic.pdf at master · grapetechvn/Grapini32_Grape32Unleashed_Boards
  - Thư viện LCD: Grapini32_Grape32Unleashed_Boards/Library/lcd at master · grapetechvn/Grapini32_Grape32Unleashed_Boards
- STM32F103RC - Mainstream Performance line, Arm Cortex-M3 MCU with 256 Kbytes of Flash memory, 72 MHz CPU, motor control, USB and CAN - STMicroelectronics

Other Development Kits

- Shops:
  - Hshop.vn - Điện tử & Robot
  - Linh kiện điện tử, điện công nghiệp, tự động hóa, cơ khí | Thegioiic
  - IC ĐÂY RỒI - THẾ GIỚI LINH KIỆN ĐIỆN TỬ



